

Polygons. Interior and exterior angles

Splitting a polygon into triangles gives $180^\circ(n - 2)$ — and the exterior angles always add to 360° , however many sides there are.

LESSON 3–4

2 × 40 MIN

GEOMETRY 8, TEMA 1

STAGE 9 · 5.2–5.3

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define a polygon, its vertices, sides, diagonals and angles.
- Distinguish convex from non-convex polygons.
- Derive and use $180^\circ(n - 2)$ for the sum of the interior angles.
- Use the fact that the exterior angles of any convex polygon add to 360° .

TEXTBOOK

National	Tema 1, pp. 5–7
Cambridge	Learner’s Book pp. 109–118
Workbook	Workbook 5.2–5.3

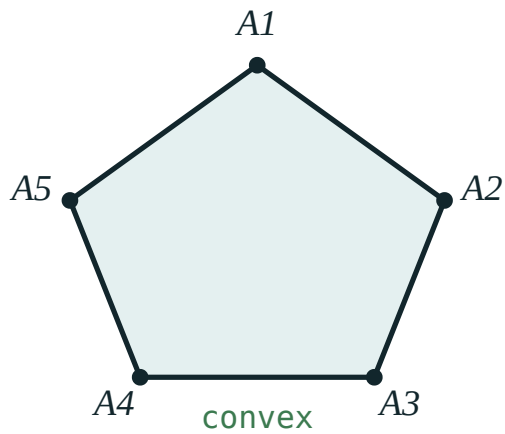
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Explanation

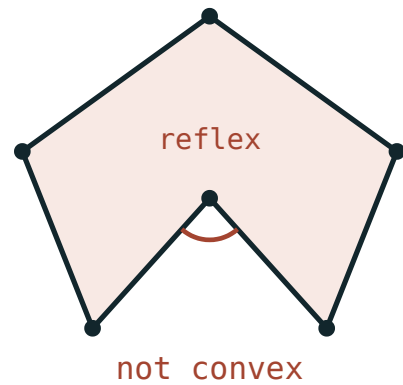
What a polygon is

DEFINITION

A **polygon** is a closed broken line together with the part of the plane it encloses. Its corners are **vertices**, its segments are **sides**, and a segment joining two non-adjacent vertices is a **diagonal**. A polygon with n sides is an n -gon and has n vertices and n angles.



Convex — the whole polygon lies on one side of every side-line.



Not convex — one interior angle is reflex, and the polygon crosses the line of a side.

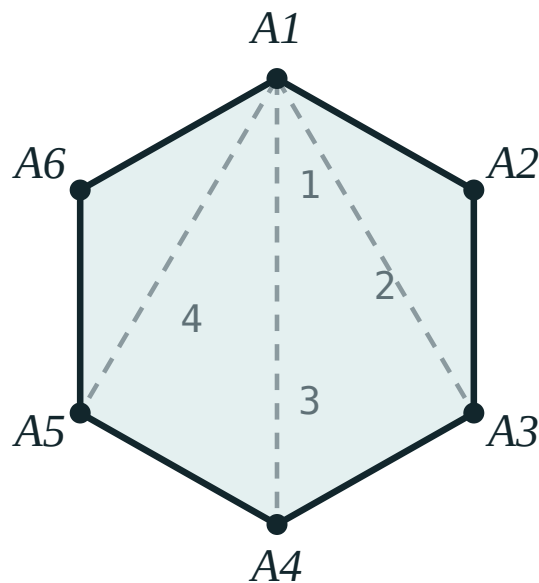
A polygon is **convex** if it lies entirely on one side of the line containing any of its sides. Equivalently: every interior angle is less than 180° . Every triangle is convex.

The sum of the interior angles

THEOREM

The interior angles of a convex n -gon add up to

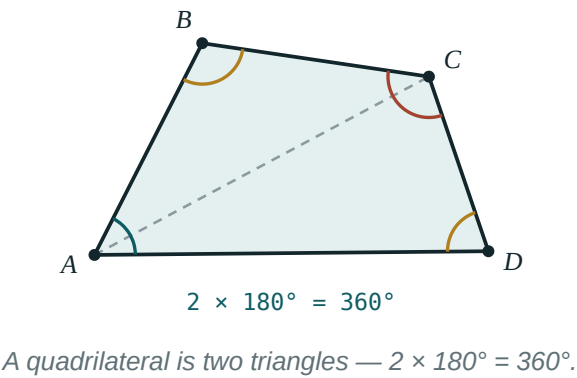
$$180^\circ \cdot (n - 2)$$



$n - 2$ triangles

*All the diagonals from one vertex split the hexagon into
 $4 = 6 - 2$ triangles.*

Proof. Choose one vertex and draw all the diagonals from it. They cut the n -gon into exactly $n - 2$ triangles — one fewer than the number of sides you can “see” from that vertex. The angles of all these triangles together make up exactly the angles of the polygon, so the sum is $180^\circ(n - 2)$. ■

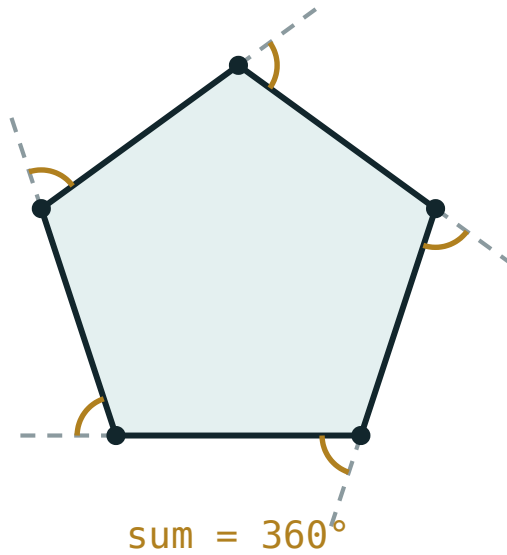


POLYGON	N	TRIANGLES	ANGLE SUM	EACH ANGLE IF REGULAR
Triangle	3	1	180°	60°
Quadrilateral	4	2	360°	90°
Pentagon	5	3	540°	108°
Hexagon	6	4	720°	120°
Octagon	8	6	1080°	135°

The exterior angles

THEOREM

Taking one exterior angle at each vertex, the exterior angles of any convex polygon add up to 360° — **whatever the number of sides.**



One exterior angle at each vertex. Walking once round the polygon turns you through a full circle.

Why. At each vertex, interior + exterior = 180° . Over n vertices that is $180^\circ n$ altogether, and the interior angles account for $180^\circ(n - 2) = 180^\circ n - 360^\circ$. What is left for the exterior angles is exactly 360° .

The walking picture is worth showing: set off along one side and walk right round the polygon. At each corner you turn by the exterior angle, and when you arrive back at the start you are facing your original direction — you have turned through one full revolution.

For a **regular** n -gon this makes the arithmetic immediate: each exterior angle is

$$\frac{360^\circ}{n}, \text{ so each interior angle is } 180^\circ - \frac{360^\circ}{n}.$$

02 Worked examples

EXAMPLE 1 Find the sum of the interior angles of a decagon, and each angle if it is regular.

① $n = 10$

A decagon has ten sides.

② $180^\circ(10 - 2) = 180^\circ \cdot 8 = 1440^\circ$

Apply the formula.

③ $\frac{1440^\circ}{10} = 144^\circ$

Regular means all ten angles are equal.

④ *check: exterior angle* $= \frac{360^\circ}{10} = 36^\circ$, and $180^\circ - 36^\circ = 144^\circ \checkmark$

Two routes, same answer.

ANSWER

1440° ; each angle 144°

EXAMPLE 2 The interior angles of a convex polygon add up to 1260° . How many sides has it?

① $180^\circ(n - 2) = 1260^\circ$

Set up the equation.

② $n - 2 = 7$

Divide by 180.

③ $n = 9$

A nonagon.

ANSWER

9 sides

EXAMPLE 3 Each interior angle of a regular polygon is 156° . Find the number of sides.

- ① $\text{exterior angle} = 180^\circ - 156^\circ = 24^\circ$
Work with the exterior angle — far quicker.

- ② $n \cdot 24^\circ = 360^\circ$
All exterior angles are equal and add to 360° .

- ③ $n = 15$

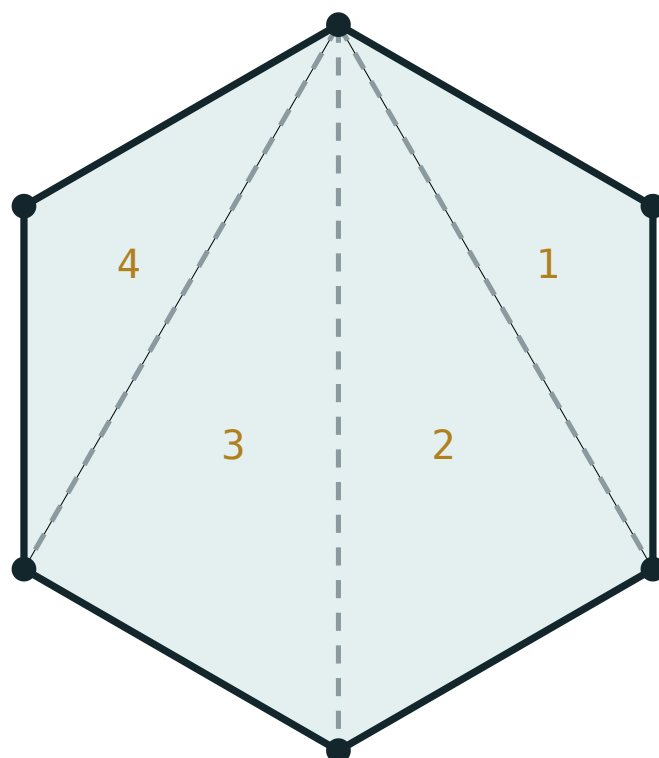
ANSWER

15 sides

03 Interactive model

Slide n from 3 to 12 and let the class read off the triangle count before the formula appears.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**

triangles **4**

interior sum **720°**

each angle **120°**

exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polygon	Ko'pburchak	Многоугольник
Convex	Qavariq	Выпуклый
Side	Tomon	Сторона
Diagonal	Diagonal	Диагональ
Interior angle	Ichki burchak	Внутренний угол
Exterior angle	Tashqi burchak	Внешний угол
Regular polygon	Muntazam ko'pburchak	Правильный многоугольник
Sum of the angles	Burchaklar yig'indisi	Сумма углов
Pentagon	Beshburchak	Пятиугольник
Hexagon	Oltiburchak	Шестиугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The interior angles of a hexagon add up to:

The exterior angles of a convex 20-gon add up to:

Each exterior angle of a regular polygon is 45° . It has:

A polygon is convex when:

all its sides are equal

every interior angle is less than 180°

it has an even number of sides

its diagonals are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Sum of the interior angles of a pentagon*

ANSWER 540°

2. *Sum of the interior angles of an octagon*

ANSWER 1080°

3. *Each interior angle of a regular hexagon*

ANSWER 120°

4. *Each exterior angle of a regular pentagon*

ANSWER 72°

5. *Sum of the exterior angles of a convex 12-gon*

ANSWER 360°

6. *How many triangles do the diagonals from one vertex of a heptagon make?*

ANSWER 5

7. *Three angles of a quadrilateral are 80° , 95° and 100° . Find the fourth.*

ANSWER 85°

Medium · the standard question

7 PROBLEMS

1. *The interior angles of a convex polygon add to 1260° . Find n .*

ANSWER $n = 9$

2. *Each interior angle of a regular polygon is 156° . Find n .*

ANSWER $n = 15$

3. *Each exterior angle of a regular polygon is 24° . Find n and the interior angle.*

ANSWER $n = 15$, interior 156°

4. *Sum of the interior angles of a regular decagon, and each angle*

ANSWER 1440° , 144°

5. *The angles of a quadrilateral are in the ratio $1 : 2 : 3 : 4$. Find them.*

ANSWER 36° , 72° , 108° , 144°

6. *A convex polygon has 20 sides. Find the sum of its interior angles.*

ANSWER 3240°

7. *How many diagonals can be drawn from one vertex of an n -gon?*

ANSWER $n - 3$

Hard · stretch and reason

7 PROBLEMS

1. Each interior angle of a regular polygon is 4 times its exterior angle. Find n .

ANSWER exterior = 36° , so $n = 10$

2. The interior angles of a convex polygon add to 2340° . Find n .

ANSWER $n = 15$

3. Show that a convex polygon can have at most three acute interior angles.

ANSWER Four or more acute angles would leave four exterior angles each above 90° , already exceeding 360° .

4. Find the total number of diagonals of an n -gon.

ANSWER $\frac{n(n-3)}{2}$ — each vertex sends $n-3$, and each diagonal is counted twice.

5. Is a regular polygon with each interior angle 155° possible? Explain.

ANSWER No — the exterior angle would be 25° and $360 \div 25 = 14.4$ is not a whole number.

6. The angles of a pentagon are 5 consecutive whole numbers of degrees. Find them.

ANSWER $106^\circ, 107^\circ, 108^\circ, 109^\circ, 110^\circ$

7. A regular polygon has interior angle 3 times that of a regular hexagon minus 180° . Find n .

ANSWER interior = $3 \cdot 120^\circ - 180^\circ = 180^\circ$ — impossible, so no such polygon exists.

07 Homework

Homework — 5 problems

Geometry 8, Tema 1, pp. 5–7. Cambridge Learner's Book 5.2–5.3 for extra practice.

- Find the sum of the interior angles of a 14-gon.
- Each interior angle of a regular polygon is 162° . Find the number of sides.
- Three angles of a quadrilateral are equal and the fourth is 120° . Find them.
- A regular polygon has exterior angle 20° . Find n and the interior angle.

5. *Draw a convex and a non-convex hexagon, and mark the reflex angle on the second.*